

Mock Final Exam - Corporate Finance

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1 Capital Budgeting

1.1 Project Background and Operational Assumptions

Company A, a state-owned plantation company, plans to invest in a new sugar processing plant project. The total initial investment required for this project is Rp 1.250.000.000.000.

The physical construction of the plant will be completed across two full years. The capital expenditures will be incurred equally over each construction year (Rp 625.000.000.000 per annum). During this construction phase, the company will already incur pre-operating expenses for salaries and land rental. The salary expenses in the first year of construction will be Rp 10.000.000.000. The company expects to adjust salary expenses annually by an inflation rate of 3,5%.

Working capital requirements are assumed to be fully integrated and encapsulated within the operating cost structures. To sustain the operational capacity of the facility, a maintenance capital expenditure equivalent to 1% of the initial capital investment (Rp 12.500.000.000) will be injected every 5 years, excluding the terminal year of the project. At the end of the project's operational horizon, the plant will be decommissioned and sold at its remaining carrying value (net of accumulated straight-line depreciation).

1.2 Revenue and Cost Projections

Sales realized in the first year of full operation are projected at Rp 550.000.000.000, with continuous positive growth expected each consecutive year. The annualized sales growth rate is expected to be 7% higher than the projected baseline inflation rate (yielding a total growth rate of 10,5%). Total sales output is measured in tonnes. The introductory sugar price is set at Rp 10.000.000 per tonne, which will adjust dynamically over time inline with inflation.

The project is structured to maintain a constant EBITDA margin of 40% across its operational lifecycle. Fixed operating costs in the first year of full production are estimated at Rp 115.000.000.000, which will grow over time with inflation. The residual operating cost components are variable and will scale proportionally with sales volume.

1.3 Financing Structure and Cost of Capital

The financing framework for the asset relies on a optimized blend of debt and equity instruments, allocated at 65% and 35% respectively. The debt portion is composed of corporate bonds issued at par value, bearing an annual coupon rate of 11,5% with a maturity of 15 years. A dedicated sinking fund will be maintained for systematic principal amortization.

The required rate of return demanded by equity holders for a comparable, fully unlevered entity is 13%. The levered cost of equity (r_e) must be computed utilizing the Modigliani-Miller Theorem with corporate tax:

$$r_e = r_u + (r_u - r_d) \times \left(\frac{D}{E} \right) \times (1 - T) \quad (1)$$

Where $r_u = 13\%$, $r_d = 11,5\%$, $\frac{D}{E} = \frac{65}{35}$, and the corporate tax rate is $T = 25\%$. Tax liabilities are waived entirely during the initial 2-year construction phase. The operational tenure of the project is defined as 30 years (excluding construction). Asset depreciation is computed via the straight-line method over the 30-year economic life. Additional capital injections do not prolong or add to the asset's original economic life.

1.4 Required Deliverables and Instructions

The implementation of this valuation model requires solving for the following components:

- a) **Financial Statements:** Construct the comprehensive projected Net Income Statement and the Projected Cash Flow Statement across the 32-year timeline.
- b) **Cost of Capital:** Calculate the Weighted Average Cost of Capital (WACC) using the formula:

$$\text{WACC} = \left(\frac{E}{V} \times r_e \right) + \left(\frac{D}{V} \times r_d \times (1 - T) \right) \quad (2)$$

- c) **Net Present Value:** Compute the Net Present Value (NPV) of the project utilizing a required discount hurdle rate set exactly 1% above the computed WACC ($r_{\text{required}} = \text{WACC} + 1\%$).
- d) **Operational Break-Even Analysis:** Determine the annualized Accounting Break-Even Quantity (Q_{AOBE}) and Cash Break-Even Quantity (Q_{CBE}) for each operating year:

$$Q_{\text{AOBE}} = \frac{\text{Fixed Costs} + \text{Depreciation}}{\text{Price per Tonne} - \text{Variable Cost per Tonne}} \quad (3)$$

- e) **Operating Leverage:** Compute the Degree of Operating Leverage (DOL) for each operating year:

$$\text{DOL} = \frac{Q \times (P - V)}{Q \times (P - V) - \text{Fixed Costs}} \quad (4)$$

- f) **Financial Optimization:** Calculate the Financial Break-Even Quantity (Q_{FBE}) for each year via iterative numeric optimization (e.g., Goal Seek or Solver) to satisfy full debt service coverage.

2 Security Market Line

Consider a risk free rate of 6,5%. The expected return and measure of systemic risk of portfolio A is 18% and 1,375. Calculate the expected return for portfolio B that the twice more volatile compared to market index. Draw the security market line and include the calculation of reward-to-risk ratio.

3 Flotation cost

Company “B” has just conducted an Initial Public Offering (IPO). Company “B” appoints Investment Bank “C” as the underwriter and agrees to a firm commitment agreement. The parameters of the offering are as follows:

- **Number of shares sold (N):** 18.000.000.000 shares
- **Proceeds to issuer (P_{issuer}):** Rp250 per share
- **Offering price (P_{offer}):** Rp289 per share
- **First-day trading price (P_{market}):** Rp300 per share
- **Direct out-of-pocket costs rate (r_{direct}):** 1% of gross proceeds
- **Indirect costs rate (r_{indirect}):** 0.25% of gross proceeds

Calculate:

a. **Total direct cost**

(Hint: Include the underwriter spread: $Spread = P_{offer} - P_{issuer}$)

b. **Total indirect cost**

(Hint: Include IPO underpricing: $Underpricing = P_{market} - P_{offer}$)

c. **Flotation cost in currency and as a percentage of total funds raised**